

EDUCATION

Carnegie Mellon University

Expected May 2027

M.S. in Software Engineering | GPA: 4.0 / 4.0

Selected Coursework: Distributed Systems, Machine Learning, Data Science for Software Engineering

University of Wisconsin–Madison

May 2024

B.S. in Computer Science & B.S. in Computer Engineering | GPA: 3.72 / 4.0

Selected Coursework: Algorithms, Operating Systems, Computer Architecture

WORK EXPERIENCE

NewsBreak

May – August 2026

Software Engineering Intern

Mountain View, CA

- Built SportsBreak's Python FastAPI / MongoDB backend from its first commit to 89 endpoints on AWS EKS, with ~1,450 fully offline tests behind a ruff / pyright / pytest gate (99.7% typed); authored 344 PRs.
- Designed an LLM news-alert pipeline that scores each article on arrival and fires only above a percentile of its topic's rolling 7-day score distribution; calibrated on 4,100+ shadow decisions, it cut filler 34% → 4%.
- Sole SwiftUI engineer; shipped [SportsBreak](#) to the App Store (~6,000 installs, 1,119 peak daily active users).
- Retired my own corroboration-based breaking-news detector after 19 hours of production data showed it was structurally 15 hours too slow (901 min p50 vs. 1.8 min pipeline latency), net –882 lines.
- Redesigned a cache that could not hit (13.2MB keyed per user) into a shared 10.9MB catalog cache, 78× smaller, sized for a 512MiB pod; replaced a 25–30 request client fan-out with one backend call.
- Took real logged search queries from 0% to 100% P@1 with deterministic entity resolution (0.4ms, 206 entities) after measuring and rejecting embeddings; shipped agentic search with hybrid RRF retrieval on top.
- Traced another team's 65% LLM failure rate via Langfuse to an unbounded reasoning loop (p95 1,197 tokens vs. 16,384 runaways); a binomial model predicted 49–66% failure vs. 65% observed.

Haddee Education

June – September 2025

Engineering Lead

San Diego, CA

- Led an 8-intern team to take a React, Next.js, FastAPI, and Supabase web app from 0→1 on GCP App Engine; designed schemas and REST APIs for LangChain resume parsing and AI job matching.

ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch

September – December 2023

Full Stack Developer

Beijing, China

- Built a VC-backed startup's LLM mock-interview web app from 0→1 as the sole engineer in React, Next.js, Supabase, and PostgreSQL; demoed it and won contracts with mid-sized companies.

RESEARCH EXPERIENCE

Control & Learning Group — Carnegie Mellon University

February – June 2026

Graduate Researcher | Advisor: Prof. Yorie Nakahira

- Developed an Online Adaptive Probabilistic Safety Certificate (PSC) with PhD researchers for stochastic systems; outperformed Adaptive MPC in 3-DOF vehicle simulations under environmental uncertainty.
- Implemented Bayesian estimators for real-time road-friction adaptation (LuGre tire models) and mapped natural-language instructions to formal safety specs with multi-turn LLMs.

SKILLS

Languages	Python, C, Go, Java, SQL, TypeScript, Swift, JavaScript, Bash
Data & ML	PyTorch, NumPy, CUDA, OpenCV, MongoDB, PostgreSQL, MySQL, RAG, hybrid retrieval
Cloud & Infra	AWS, GCP, Kubernetes (EKS), Docker, Helm, Jenkins, GitHub Actions, CI/CD, Linux
Frameworks	FastAPI, Pydantic, LangChain, React, Next.js, Supabase
Practices	Git, Unit Testing, pytest, pyright, ruff, TDD, REST APIs, Agile/Scrum

PROJECTS

- **my-av**: Built a PyTorch CNN-GRU that predicts steering and speed from 20-frame (1s) video; built a data pipeline over 33 hours of Comma2k19 and ran streaming inference on local CUDA and GCP Vertex AI.
- **Wisconsin Autonomous**: Integrated wheel-speed encoders on the competition rover and streamed real-time data to the onboard NVIDIA Jetson.